Q1 - 2024 (01 Feb Shift 1)

Let$S=\{z \in C:|z-1|=1$ and $(\sqrt{2}-1)(z+\bar{z})-i(z-\bar{z})=2 \sqrt{2}\}$. Let $z_1, \quad z_2 \in S$ be such that $|z_1|=\max_{z \in s}|z|$ and $|z_2|=\min_{z \in S}|z|$. Then $\left|\sqrt{2} z_1-z_2\right|^2$ equals :

(1) 1

(2) 4

(3) 3

(4) 2

Q2 - 2024 (01 Feb Shift 1)

Let $\mathrm{P}=\{\mathrm{z} \in \mathbb{C}:|\mathrm{z}+2-3 \mathrm{i}| \leq 1\}$ and $Q=\{z \in \mathbb{C}: z(1+i)+\bar{z}(1-i) \leq-8\}$. Let in $\mathrm{P} \cap \mathrm{Q}$, $|\mathrm{z}-3+2 \mathrm{i}|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2+2|z|^2=\alpha+\beta \sqrt{2}$, where α, β are integers, then $\alpha+\beta$ equals

Q3 - 2024 (01 Feb Shift 2)

If z is a complex number such that $|z| \geq 1$, then the minimum value of $\left|\mathrm{z}+\frac{1}{2}(3+4 i)\right|$ is:

[We changed options. In official NTA paper no option was correct.]

(1) $\frac{5}{2}$

(2) 2

(3) 3

(4) 0

Q4 - 2024 (27 Jan Shift 1)

If $S=\{z \in C:|z-i|=|z+i|=|z-1|\}$, then, $n(S)$ is:

(1) 1

(2) 0

(3) 3

(4) 2

Q5 - 2024 (27 Jan Shift 1)

If α satisfies the equation $x^2+x+1=0$ and $(1+\alpha)^7=\mathrm{A}+\mathrm{B}\alpha+\mathrm{C}^2,\ \mathrm{A},\ \mathrm{B}, \mathrm{C} \geq 0$, then $5(3\ \mathrm{A}-2\ \mathrm{B}-\mathrm{C})$ is equal to_____

Q6 - 2024 (27 Jan Shift 2)

Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circles $\left|\mathrm{z}-\mathrm{z}_0\right|^2=4$ and $\left|\mathrm{z}-\mathrm{z}_0\right|^2=16$ respectively, where $\mathrm{z}_0=1+\mathrm{i}$. Then, the value of $100|\alpha|^2$ is.

Q7 - 2024 (29 Jan Shift 1)

If $z=\frac{1}{2}-2i$, is such that $|z+1|=\alpha z+\beta(1+i), i=\sqrt{-1}$ and $\alpha, \beta \in R$, then $\alpha+\beta$ is equal to

(1) -4

(2) 3

(3) 2

(4) -1

Q8 - 2024 (29 Jan Shift 1)

Let α, β be the roots of the equation $x^2-x+2=0$ with $\operatorname{Im}(\alpha)>\operatorname{Im}(\beta)$. Then $\alpha^6+\alpha^4+\beta^4-5\alpha^2$ is equal to

Q9 - 2024 (29 Jan Shift 2)

Let r and θ respectively be the modulus and amplitude of the complex number $z=2-i\left(2\tan\frac{5\pi}{8}\right)$, then (r, θ) is equal to

(1) $\left(2\sec\frac{3\pi}{8}, \frac{3\pi}{8}\right)$

(2) $\left(2\sec\frac{3\pi}{8}, \frac{5\pi}{8}\right)$

(3) $\left(2\sec\frac{5\pi}{8}, \frac{3\pi}{8}\right)$

(4) $\left(2\sec\frac{11\pi}{8}, \frac{11\pi}{8}\right)$

Q10 - 2024 (29 Jan Shift 2)

Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\operatorname{Im}(\alpha) > \operatorname{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^{\mathrm{n}}(\mathrm{a} + \mathrm{ib}), \mathrm{i} = \sqrt{-1}$. Then $\mathrm{n} + \mathrm{a} + \mathrm{b}$ is equal to ______.

Q11 - 2024 (30 Jan Shift 1)

If $z = x + iy, xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $|z^2|$ is equal to :

(1) 9

(2) 1

(3) 4

(4) $\frac{1}{4}$

Q12 - 2024 (30 Jan Shift 2)

If z is a complex number, then the number of common roots of the equation $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$, is equal to :

(1) 1

(2) 2

(3) 0

(4) 3

Q13 - 2024 (31 Jan Shift 1)

If α denotes the number of solutions of $|1-i|^x = 2^x$ and $\beta = \left(\frac{|z|}{\arg(z)}\right)$, where $z = \frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}\mathrm{i}}{\sqrt{\pi}+\mathrm{i}} + \frac{\sqrt{\pi}-\mathrm{i}}{1+\sqrt{\pi}\mathrm{i}}\right), \mathrm{i} = \sqrt{-1}$, then the distance of the point (α, β) from the line $4x - 3y = 7$ is ______.

Q14 - 2024 (31 Jan Shift 2)

Let z_1 and z_2 be two complex number such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$. Then $|z_1^4 + z_2^4|$ equals-

(1) $30\sqrt{3}$

(2) 75

(3) $15\sqrt{15}$

(4) $25\sqrt{3}$

Answer Key

Q1 (4)	**Q2** (36)	**Q3** (4)	**Q4** (1)
Q5 (5)	**Q6** (20)	**Q7** (2)	**Q8** (13)
Q9 (1)	**Q10** (49)	**Q11** (2)	**Q12** (2)
Q13 (3)	**Q14** (2)		

Q1

Let $Z = x +$ iy

Then $(\mathrm{x}-1)^2+\mathrm{y}^2=1 \quad \rightarrow [OPTION1]$

& $\quad (\sqrt{2}-1)(2x)-i(2iy)=2\sqrt{2}$

$\Rightarrow (\sqrt{2}-1)x+y=\sqrt{2} \rightarrow (2)$

Solving (1) & (2)

we get

Either $\mathrm{x}=1$ or $x=\frac{1}{2-\sqrt{2}} \rightarrow (3)$

On solving (3) with (2)

we get

For $\mathrm{x}=1 \Rightarrow \mathrm{y}=1 \Rightarrow \mathrm{Z}_2=1+\mathrm{i}$ & for

$x=\frac{1}{2-\sqrt{2}} \Rightarrow y=\sqrt{2}-\frac{1}{\sqrt{2}} \Rightarrow Z_1=\left(1+\frac{1}{\sqrt{2}}\right)+\frac{i}{\sqrt{2}}$

Now

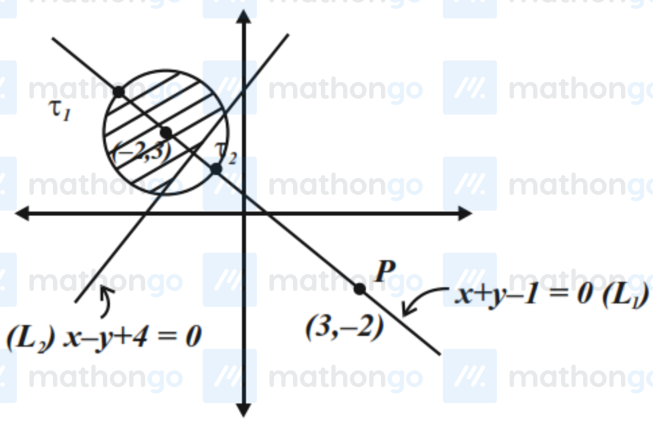
$\left|\sqrt{2}z_1-z_2\right|^2$

$=\left|\left(\frac{1}{\sqrt{2}}+1\right)\sqrt{2}+i-(1+i)\right|^2$

$=(\sqrt{2})^2$

$=2$

Q2



Clearly for the shaded region z_1 is the intersection of the circle and the line passing through P (L_1) and z_2 is intersection of line L_1& L_2

Circle: $(x+2)^2+(y-3)^2=1$

$L_1 : x+y-1=0$

$L_2 : x-y+4=0$

On solving circle & L_1 we get

$z_1 : \left(-2-\frac{1}{\sqrt{2}}, 3+\frac{1}{\sqrt{2}}\right)$

On solving L_1 and z_2 is intersection of line L_1 & L_2

we get $z_2 : \left(\frac{-3}{2}, \frac{5}{2}\right)$

$$|z_1|^2+2|z_2|^2=14+5\sqrt{2}+17$$

$$=31+5\sqrt{2}$$

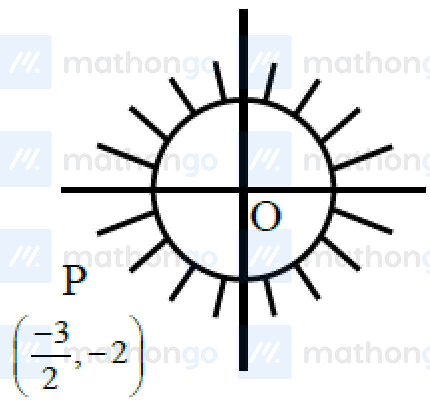
$\alpha=31$

So $\beta=5$

$\alpha+\beta=36$

Q3

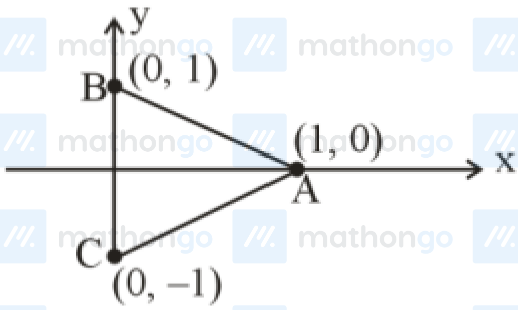
$|z| \geq 1$



Min. value of $\left|z+\frac{3}{2}+2i\right|$ is actually zero.

Q4

$|z-i|=|z+i|=|z-1|$



ABC is a triangle. Hence its circum-centre will be the only point whose distance from A, B, C will be same.

So $\mathrm{n}(\mathrm{S})=1$

Q5

$x^2+x+1=0 \Rightarrow x=\omega, \omega^2=\alpha$

Let $\alpha=\omega$

Now $(1+\alpha)^7=-\omega^{14}=-\omega^2=1+\omega$

$\mathrm{A}=1,\ \mathrm{B}=1, \mathrm{C}=0$

$\therefore 5(3A-2B-C)=5(3-2-0)=5$

Q6

$$\begin{aligned}
& |z-z_0|^2=4 \\
& \Rightarrow (\alpha-z_0)(\bar{\alpha}-\bar{z}_0)=4 \\
& \Rightarrow \alpha\bar{\alpha}-\alpha\bar{z}_0-z_0\bar{\alpha}+|z_0|^2=4 \\
& \Rightarrow |\alpha|^2-\alpha\bar{z}_0-z_0\bar{\alpha}=2 \ldots\ldots\ldots(1) \\
& |z-z_0|^2=16 \\
& \Rightarrow \left(\frac{1}{\bar{\alpha}}-z_0\right)\left(\frac{1}{\alpha}-\bar{z}_0\right)=16 \\
& \Rightarrow (1-\bar{\alpha}z_0)(1-\alpha\bar{z}_0)=16|\alpha|^2 \\
& \Rightarrow 1-\bar{\alpha}z_0-\alpha\bar{z}_0+|\alpha|^2|z_0|^2=16|\alpha|^2 \\
& \Rightarrow 1-\bar{\alpha}z_0-\alpha\bar{z}_0=14|\alpha|^2 \ldots\ldots\ldots\ldots(2)
\end{aligned}$$

From (1) and (2)

$$\begin{aligned}
& \Rightarrow 5|\alpha|^2=1 \\
& \Rightarrow 100|\alpha|^2=20
\end{aligned}$$

Q7

$z=\frac{1}{2}-2i$

$|z+1|=\alpha z+\beta(1+i)$

$\left|\frac{3}{2}-2i\right|=\frac{\alpha}{2}-2\alpha i+\beta+\beta i$

$\left|\frac{3}{2}-2i\right|=\left(\frac{\alpha}{2}+\beta\right)+(\beta-2\alpha)i$

$\beta=2\alpha$ and $\frac{\alpha}{2}+\beta=\sqrt{\frac{9}{4}+4}$

$\alpha+\beta=3$

Q8

$\alpha^6+\alpha^4+\beta^4-5\alpha^2$

$=\alpha^4(\alpha-2)+\alpha^4-5\alpha^2+(\beta-2)^2$

$=\alpha^5-\alpha^4-5\alpha^2+\beta^2-4\beta+4$

$=\alpha^3(\alpha-2)-\alpha^4-5\alpha^2+\beta-2-4\beta+4$

$=-2\alpha^3-5\alpha^2-3\beta+2$

$=-2\alpha(\alpha-2)-5\alpha^2-3\beta+2$

$=-7\alpha^2+4\alpha-3\beta+2$

$=-7(\alpha-2)+4\alpha-3\beta+2$

$=-3\alpha-3\beta+16=-3(1)+16=13$

Q9

$z=2-i\left(2\tan\frac{5\pi}{8}\right)=x+iy(\text{ let })$

$r=\sqrt{x^2+y^2} \& \theta=\tan^{-1}\frac{y}{x}$

$r=\sqrt{(2)^2+\left(2\tan\frac{5\pi}{8}\right)^2}$

$=\left|2\sec\frac{5\pi}{8}\right|=\left|2\sec\left(\pi-\frac{3\pi}{8}\right)\right|$

$=2\sec\frac{3\pi}{8}$

$\&\theta=\tan^{-1}\left(\frac{-2\tan\frac{5\pi}{8}}{2}\right)$

$=\tan^{-1}\left(\tan\left(\pi-\frac{5\pi}{8}\right)\right)$

$=\frac{3\pi}{8}$

Q10

$x^2-\sqrt{6}x+6=0_{\Delta\beta}^{\alpha}$

$x=\frac{\sqrt{6}\pm i\sqrt{6}}{2}=\frac{\sqrt{6}}{2}(1\pm i)$

$\alpha=\sqrt{3}\left(e^{i\frac{\pi}{4}}\right),\beta=\sqrt{3}\left(e^{-i\frac{\pi}{4}}\right)$

$\therefore\frac{\alpha^{99}}{\beta}+\alpha^{98}=\alpha^{98}\left(\frac{\alpha}{\beta}+1\right)$

$=\frac{\alpha^{98}(\alpha+\beta)}{\beta}=3^{49}\left(e^{i99\frac{\pi}{4}}\right)\times\sqrt{2}$

$=3^{49}(-1+\mathrm{i})$

$=3^{\mathrm{n}}(\mathrm{a}+\mathrm{ib})$

$\therefore\mathrm{n}=49,\mathrm{a}=-1,\ \mathrm{b}=1$

$\therefore\mathrm{n}+\mathrm{a}+\mathrm{b}=49-1+1=49$

Q11

$\mathrm{z}^2=-\mathrm{i}\bar{\mathrm{z}}$

$\left|\mathrm{z}^2\right|=|\mathrm{iz}|$

$\left|z^2\right|=|z|$

$|z|^2-|z|=0$

$|z|(|z|-1)=0$

$|z|=0$ (not acceptable)

$\therefore |z| = 1$

$\therefore |z|^2 = 1$

Q12

Sol. $z^{1985} + z^{100} + 1 = 0 \& z^3 + 2z^2 + 2z + 1 = 0$

$(z+1)\left(z^2 - z + 1\right) + 2z(z+1) = 0$

$(z+1)\left(z^2 + z + 1\right) = 0$

$\Rightarrow \quad z = -1, \quad z = w, w^2$

Now putting $z = -1$ not satisfy

Now put $\mathrm{z} = \mathrm{w}$

$\Rightarrow \quad \mathrm{w}^{1985} + \mathrm{w}^{100} + 1$

$\Rightarrow \quad \mathrm{w}^2 + \mathrm{w} + 1 = 0$

Also, $\mathrm{z} = \mathrm{w}^2$

$\Rightarrow \quad \mathrm{w}^{3970} + \mathrm{w}^{200} + 1$

$\Rightarrow \quad \mathrm{w} + \mathrm{w}^2 + 1 = 0$

Two common root

Q13

$(\sqrt{2})^x = 2^x \Rightarrow x = 0 \Rightarrow \alpha = 1$

$$z = \frac{\pi}{4}(1+i)^4\left[\frac{\sqrt{\pi} - \pi i - i - \sqrt{\pi}}{\pi + 1} + \frac{\sqrt{\pi} - i - \pi i - \sqrt{\pi}}{1 + \pi}\right]$$

$$= -\frac{\pi i}{2}\left(1 + 4i + 6i^2 + 4i^3 + 1\right)$$

$$= 2\pi i$$

$$\beta = \frac{2\pi}{\frac{\pi}{2}} = 4$$

Distance from $(1,4)$ to $4x - 3y = 7$

Will be $\frac{15}{5} = 3$

Q14

$z_1 + z_2 = 5$
$z_1^3 + z_2^3 = 20 + 15i$
$z_1^3 + z_2^3 = (z_1 + z_2)^3 - 3z_1z_2(z_1 + z_2)$
$z_1^3 + z_2^3 = 125 - 3z_1 \cdot z_2(5)$
$\Rightarrow 20 + 15i = 125 - 15z_1z_2$
$\Rightarrow 3z_1z_2 = 25 - 4 - 3i$
$\Rightarrow 3z_1z_2 = 21 - 3i$
$\Rightarrow z_1 \cdot z_2 = 7 - i$
$\Rightarrow (z_1 + z_2)^2 = 25$
$\Rightarrow z_1^2 + z_2^2 = 25 - 2(7 - i)$
$\Rightarrow 11 + 2i$
$(z_1^2 + z_2^2)^2 = 121 - 4 + 44i$
$\Rightarrow z_1^4 + z_2^4 + 2(7 - i)^2 = 117 + 44i$
$\Rightarrow z_1^4 + z_2^4 = 117 + 44i - 2(49 - 1 - 14i)$
$\Rightarrow |z_1^4 + z_2^4| = 75$